Sahaj Bhadja

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Summary

Software engineer pursuing an MSCS with 2+ years building event-driven integrations, backend services, ETL workflows, and production validation tooling using Python, JavaScript/TypeScript, SQL, REST APIs, and cloud platforms.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built **ETL-style backend workflows** for 500+ users, translating stakeholder requirements into APIs, validation logic, schema updates, and maintainable release pipelines across structured data systems and production-facing integrations.
- Improved **performance and operational reliability** by tuning indexed queries, background jobs, and CI/CD releases, reducing dashboard latency from **1.2s to 350ms** while strengthening debugging, traceability, and validation workflows.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed **Python services and JSON-backed workflows** handling 10,000+ daily requests, integrating uploads, OCR processing, persistence logic, and model-evaluation pipelines while supporting feature validation and production-ready delivery for instructors.
- Created **debugging notes, QA checks, and issue fixes** across upload, OCR, persistence, and grading features, reducing feedback turnaround by **60%** while improving observability, consistency, and release confidence.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered **cross-platform APIs and ETL integrations** using Python, SQL, AWS Glue, and S3, improving response time by **30%** while supporting data movement, transformation workflows, and reusable implementation patterns.
- Maintained **20+ automated validation suites** with Jenkins, PyTest, and Mockito, increasing CI reliability by **68%** and strengthening production support through repeatable test coverage, defect isolation, and release documentation.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built **REST and WebSocket modules** over SQL/NoSQL persistence, supporting 2M+ synchronized records across operational workflows while improving responsiveness, real-time data exchange, and dependable full-stack integration behavior in production.
- Automated **Docker-based build and test workflows** with Jenkins, cutting release time by **50%** while improving deployment consistency, troubleshooting speed, validation coverage, and engineering handoffs across iterative feature delivery.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built **Flask APIs and event-triggered notification flows** using Firebase Cloud Messaging for 50K+ records, supporting webhook integrations, token-based access control, and documented client-facing release workflows across web systems.
- Integrated **payment and notification features** with maintainable backend logic and technical documentation, reducing manual handling by **70%** while improving operational coordination, usability, and production support readiness overall.

Skills

Languages: Python, JavaScript, TypeScript, SQL, Java, C++, Bash, JSON
APIs & Integrations: REST APIs, WebSockets, webhook integrations, event-driven workflows, payload validation, technical documentation
Data & QA: SQL querying, ETL workflows, data modeling, automated checks, PyTest, Mockito, debugging, runbook-style validation
Platforms & Tools: Flask, React, Node.js, AWS Glue, S3, PostgreSQL, MySQL, MongoDB, Docker, Jenkins, Git/GitHub, CI/CD, Agile

Projects

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB, Plaid*
- Built a **real-time fintech platform** supporting 50K+ users and 1M+ transactions with secure bank linking, event-driven updates, and filtered transaction views across scalable TypeScript-backed workflows and infrastructure.

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP*
- Built an **AI-powered document platform** with secure upload, semantic search, summarization, and structured data handling over 50K+ records, supporting low-latency retrieval and reusable ingestion patterns for analysis workflows.

HMIS Live | *MERN, ABDM APIs, Twilio, Sentry, TypeScript*
- Built a **healthcare operations platform** with consent-based access, appointment workflows, notifications, and monitoring, reducing administrative workload by **30%** while improving timely data access and coordination across teams.